#HW 5

Graded

Student

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Total Points

92 / 100 pts

Question 1

1 — 8 / 10 pts

- – 0 pts Correct
- ✔ – 2 pts You should find out the compressibility factor in both cases
- – 2 pts Compressibility factor calculation is not correct.

Question 2

2 — 10 / 10 pts

- ✔ – 0 pts Correct
- – 2 pts The temperature has been calculated incorrectly.
- – 2 pts The work calculation is wrong.
- – 3 pts incomplete

Question 3

3 — 13 / 15 pts

- – 0 pts Correct
- – 2 pts Heat transfer had to be calculated in kJ.
- – 2 pts Work calculation is wrong.
- ✔ – 2 pts The heat transfer calculation is wrong. Look into your del(U) calculation.
- – 3 pts incomplete

Question 4

4 — 11 / 15 pts

- – 0 pts Correct
- – 2 pts W is correct, but Q is wrong. Del(U) will not be 0 here.
- ✔ – 4 pts The process is right, but the calculation is wrong.
- – 2 pts The heat transfer calculation is wrong.

Question 5

5 **15** / 15 pts

✔ **– 0 pts** Correct

– 2 pts W12 at constant pressure has to be calculated through PdV.

– 2 pts You have not calculated Q31 and W31

– 4 pts Your calculations are wrong.

– 6 pts incomplete

– 10 pts Late

Question 6

6 **35** / 35 pts

✔ **– 0 pts** Correct

No questions assigned to the following page.

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Question #1

EXERCISE | 3.21.67: (Problem 3.67 in the 9th edition).

Check the applicability of the ideal gas model for

(a) water at 600°F and pressures of 900 lbf/in.^{2} and 100 lbf/in.^{2}

(b) nitrogen at −20°C and pressures of 75 bar and 1 bar.

Feedback?

a_1) 600°F , 900 lbf/in^2 (H_2O)

$$°C = \frac{5(°F) - 32)}{9} = \frac{5(600-32)}{9} = 315.56\,°C + 273.15 = 588.71\,K$$

$$\frac{900\,lbf}{in^2} \Rightarrow 62.0528\,bar$$

$T_{c_{H_2O}} = 647.3\,K$

$P_{c_{H_2O}} = 220.9\,bar$

$$T_R = \frac{T}{T_c} = \frac{588.71}{647.3} = 0.909$$

$$P_R = \frac{P}{P_c} = \frac{62.0528}{220.9} = 0.281$$

a_2) 600°F , 100 lbf/in^2 (H_2O)

600°F ⇒ 588.71 K

100 lbf/in^2 ⇒ 6.89 bar

$T_{c_{H_2O}} = 647.3\,K$

$P_{c_{H_2O}} = 220.9\,bar$

$$T_R = \frac{T}{T_c} = \frac{588.71}{647.3} = 0.909$$

$$P_R = \frac{P}{P_c} = \frac{6.89}{220.9} = 0.031$$

b_1) −20°C , 75 bar (N_2)

−20°C + 273.15 = 253.15 K

$T_{c_{N_2}}$: 126 K

$P_{c_{N_2}}$: 33.9 bar

not applicable; both pressure & temp. are above critical values.

b_2) −20°C , 1 bar (N_2)

−20°C = 253.15 K

1 bar

not applicable ; pressure is below critical value but the temperature is still above critical value.

No questions assigned to the following page.

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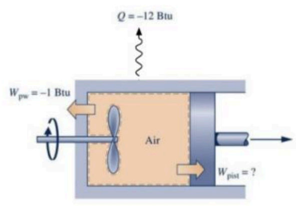
Question #2

EXERCISE 3.21.74: (Problem 3.74 in the 9th edition).

(a) As shown in the figure below, a piston–cylinder assembly fitted with a paddle wheel contains air, initially at p_1 = 30 lbf/in.², T_1 = 540°F, and V_1 = 4 ft³. The air undergoes a process to a final state where p_2 = 20 lbf/in.², V_2 = 4.5 ft³. During the process, the paddle wheel transfers energy *to* the air by work in the amount of 1 Btu, and there is energy transfer *from* the air by heat in the amount of 12 Btu. Assuming ideal gas behavior, and neglecting kinetic and potential energy effects, determine for the air (a) the temperature at state 2, in °R, and (b) the energy transfer by work from the air to the piston, in Btu.



Initially, p_1 = 30 lbf/in.², T_1 = 540°F, V_1 = 4 ft³.
Finally, p_2 = 20 lbf/in.², V_2 = 4.5 ft³.

Feedback?

a)

$PV = RT \Rightarrow \frac{P_1V_1}{R_1T_1} = \frac{P_2V_2}{R_2T_2}$

$30 \frac{\text{lbf}}{\cancel{\text{in}^2}} \cdot \frac{12\cancel{\text{in}}}{1\text{ft}} \cdot \frac{12\cancel{\text{in}}}{1\text{ft}} = \underline{4320\ \text{lbf/ft}^2}$

→ p_1 = 30 lbf/in²
→ T_1 = 540°F
→ V_1 = 4 ft³

initial state

°F + 459.67 = °R
540°F + 459.67 = $\underline{999.67°R}$

→ p_1 = 20 lbf/in²
→ V_1 = 4.5 ft³

final state

expanding (work is negative)

$$\frac{(4320\ \text{lbf/ft}^2)(4\ \text{ft}^3)}{(1545\ \text{ft lbf/lbmol°R})(999.67°R)} = \frac{(2880\ \text{lbf/ft}^2)(4.5\ \text{ft}^3)}{(1545\ \text{ft lbf/lbmol°R})(T_2)}$$

$20\ \text{lbf/in}^2 \cdot \frac{12\text{in}}{1\text{ft}} \cdot \frac{12\text{in}}{1\text{ft}} = \underline{2880\ \text{lbf/ft}^2}$

$T_2\ 0.011188 = \frac{12960}{(1545)\cancel{T_2}} \cdot \cancel{T_2}$

$T_2 = 749.76°R$

b) $\cancel{\Delta KE}^{0} + \cancel{\Delta PE}^{0} + \Delta U = Q - W$

$\Delta U = Q - (W_{pad.} + W_{pist.})$; $Q = -12$ Btu, $W_{pad} = -1$ Btu ; $\Delta U = mc_v(T_2 - T_1)$

$P_1V_1 = mRT_1 \Rightarrow m = \frac{P_1V_1}{RT_1}$

$m = \frac{(30\ \text{psia})(4\ \text{ft}^3)}{(0.3704\ \text{Btu/lbm°R})(999.67°R)} = 0.3243$

$\Delta U = (0.3243)(0.171)(749.75 - 999.67) = -13.85$ Btu

-13.85 Btu $= -12$ Btu $+ 1$ Btu $+ W_{piston}$

$W_{piston} = -2.85$ Btu

No questions assigned to the following page.

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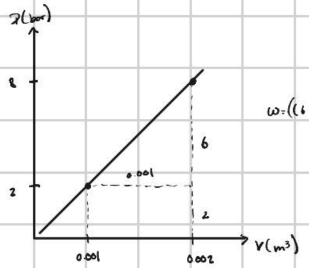
Question # 3

EXERCISE | 3.21.76: (Problem 3.76 in the 9th edition).

(a) Air contained in a piston–cylinder assembly, initially at 2 bar, 200 K, and a volume of 1 L, undergoes a process to a final state where the pressure is 8 bar and the volume is 2 L. During the process, the pressure–volume relationship is linear. Assuming the ideal gas model for the air, determine the work and heat transfer, each in kJ.

Feedback?

a)



$$W = \left((6 \cdot 0.001 \cdot 0.5) + (2 \cdot 0.001)\right) = 5 \times 10^{-3}\ \text{bar} \cdot m^3 \cdot \frac{10^5\ Pa}{bar} = 500\ \text{Joule} \cdot \frac{1\ kJ}{1000\ J} = 0.50\ kJ$$

Work

b)

$$P v = R T \quad v = \frac{V}{m} \Rightarrow \frac{PV}{m} = RT \Rightarrow m = \frac{P_1 V_1}{R T_1}$$

$$P_1 = 2\ bar \cdot \frac{10^5\ Pa}{1\ bar} \cdot \frac{1\ kPa}{10^3\ Pa} = 200\ kPa$$

$$T_1 = 200K$$

$$V_1 = 1L \cdot \frac{1\ m^3}{1000\ L} = 0.001\ m^3$$

$$R = 8.314\ kJ/kmol\,K$$

$$M_{air} = 17.03\ kg/kmol$$

$$\frac{(200\ \cancel{kPa})(0.001\ m^3)\left(17.03\ \frac{kg}{\cancel{kmol}}\right)}{(8.314\ \cancel{kJ/kmol\,K})(200K)} = 2.048 \times 10^{-5}\ kg \Longrightarrow m = \frac{PV}{RT} \Rightarrow T_2 = \frac{P_2 V_2}{mR}$$

$$T_2 = \frac{(800\ \cancel{kPa})(0.002\ m^3)\left(17.03\ \frac{\cancel{kg}}{\cancel{kmol}}\right)}{(2.048 \times 10^{-3}\ \cancel{kg})(8.314\ \cancel{kJ/kmol\,K})} = 1600.28\ K$$

$$v_1 = \frac{V_1}{m} = \frac{(0.001\ m^3)}{(2.048 \times 10^{-3}\ kg)} = 0.488\ m^3/kg$$

$$\cancel{\Delta KE} + \cancel{\Delta PE} + \Delta U = Q - W$$

$$0 \qquad 0$$

$$\Rightarrow \Delta U = Q - W \ ; \ \Delta U = m c_v (T_2 - T_1)$$

$$m c_v (T_2 - T_1) = Q - W$$

$$\Rightarrow Q = (m c_v (T_2 - T_1)) + W$$

$$= \left((2.048 \times 10^{-3}\ \cancel{kg})(0.718\ kJ/\cancel{kg\,K})(1600.28 - 200)\cancel{K}\right) + 0.5\ kJ$$

$$Q = 2.56\ kJ$$

No questions assigned to the following page.

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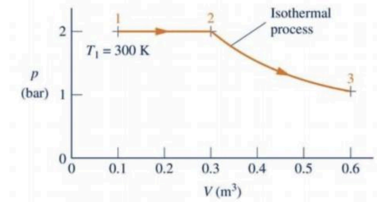
Question # 4

EXERCISE 3.21.93: (Problem 3.93 in the 9th edition).

(a) Air contained in a piston–cylinder assembly undergoes two processes in series, as shown in the figure below. Assuming ideal gas behavior for the air, determine the work and heat transfer for the overall process, each in kJ/kg.



Feedback?

a) Work for overall process

$W_{Total} = W_{1-2} + W_{2-3}$

$W = \int_{V_1}^{V_2} P\,dV \Rightarrow W_{1-2} = P\int_{V_1}^{V_2} dV \Rightarrow P[V_2 - V_1] = 2[0.3-0.1] = 0.4\ \text{bar}\cdot m^3 \cdot \frac{10^5\ Pa}{bar} = 4\times10^4\ Pa\cdot m^3 = 4\times10^4\ J \cdot \frac{1kJ}{1000J} = 40\ kJ$

Isothermal Process ; $W = nRT\ln(V_f/V_i)$; $PV = nRT$ $\Big\}$ $P\cdot V \ln(V_f/V_i) = 0.6\ln(0.6/0.3) = 0.416\ \text{bar}\cdot m^3 \cdot \frac{10^5\ Pa}{bar} = 4.16\times10^4\ Pa\cdot m^3 = 4.16\times10^4\ J\cdot\frac{1kJ}{1000J} = 41.6\ kJ$

$W = nRT\ln(P_i/P_f)$; $P\cdot V\ln(P_i/P_f) = 0.6\ln(2/1) = 0.416\ \text{bar}\cdot m^3$

$S_2 : P\cdot V = (2)(0.3) = 0.6$

$S_3 : P\cdot V = (0.6)(1) = 0.6$

$W_{Total} = 40kJ + 41.6kJ = 81.6kJ$

$m = (14.033)(28.97) = 406.536\ g \Rightarrow 0.4065\ kg$

$\frac{W}{m} = \frac{81.6kJ}{0.4065kg} = 200.74\ kJ/kg$

b) $Q_{Total} = Q_{1-2} + Q_{2-3}$

$Q = nC_p\Delta T = \cancel{n}C_p\frac{P(V_2-V_1)}{\cancel{n}R} = \frac{C_p}{R}P_1(V_2-V_1)$; for air, $C_p = \frac{7R}{2}$

$\Rightarrow \frac{7\cancel{R}}{2}\frac{1}{\cancel{R}}P_1(V_2-V_1) \Rightarrow \frac{7}{2}P_1(V_2-V_1)$

Q_{1-2} : $Q = \frac{7}{2}P_1(V_2-V_1) = \frac{7}{2}(2\ bar)(0.3-0.1)\ m^3 = 1.4\ \text{bar}\cdot m^3\ \frac{10^5\ Pa}{bar} = 1.4\times10^5\ Pa\ m^3 = 1.4\times10^5\ J\cdot\frac{1kJ}{1000J} = 140kJ$

Q_{2-3} : For isothermal processes, $\Delta U = 0$

$\underset{0}{\cancel{\Delta KE}} + \underset{0}{\cancel{\Delta PE}} + \Delta U = Q - W$

$\Delta U = Q - W$; $\Delta U = 0$

$\Rightarrow 0 = Q - W \Rightarrow W = Q \Rightarrow W_{2-3} = Q_{2-3}$

$W_{2-3} = 41.6\ kJ = Q_{2-3}$

$Q_{Total} = Q_{1-2} + Q_{2-3} = 140kJ + 41.6kJ = 181.6kJ$

$\frac{Q}{m} = \frac{181.6kJ}{0.4065kg} = 446.74\frac{kJ}{kg}$

No questions assigned to the following page.

Hw # 5

Question # 5

EXERCISE 3.21.95: (Problem 3.95 in the 9th edition).

(a) One lb of air undergoes a cycle consisting of the following processes:

- **Process 1–2:** Constant-pressure expansion with p = 20 lbf/in.^{2} from T_1 = 500°R to v_2 = 1.4 v_1.
- **Process 2–3:** Adiabatic compression to $v_3 = v_1$ and T_3 = 820°R.
- **Process 3–1:** Constant-volume process.

Sketch the cycle on a carefully labeled $p-v$ diagram. Assuming ideal gas behavior, determine the energy transfers by heat and work for each process, in Btu.

Feedback?

1 lb of Air

knowns:

$P_1 = 20\,lbf/in^2 = p_2$

$T_1 = 500°R$

$v_2 = 1.4 v_1$

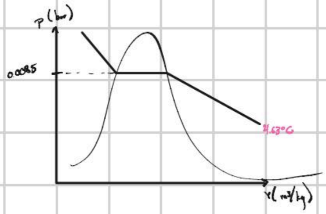
$v_3 = v_1$

$T_3 = 820°R$

$\rightarrow 20\,lbf/in^2 = \underline{1.38\ bar}$

$\rightarrow 500°R = \underline{4.63°C}$

$(°R - 491.67) \times 5/9$



Approximate

$$\frac{(0.00872 - 0.00813)\,bar}{(5-4)°C} = \frac{p - 0.00813\,bar}{(4.63-4)°C}$$

$p = 0.0085017\ bar$

Assuming ideal gas

$P \cdot v = RT \; ; \; \Delta P = 0 \; , \; R = \text{constant}$

$$\frac{v_2}{v_1} = \frac{T_2}{T_1} \Longrightarrow \frac{1.4\cancel{v_1}}{\cancel{v_1}} = \frac{T_2}{500°R} \Longrightarrow \underline{T_2 = 700°R}$$

$$W_{1-2} = m\underbrace{R}_{0.06855\ \frac{Btu}{lb°R}} \times (T_2 - T_1) = (1)(0.06855)(700-500) = \underline{13.71\ Btu}$$

$$\underbrace{\cancel{\Delta KE}}_{0} + \underbrace{\cancel{\Delta PE}}_{0} + \Delta U = Q - W$$

$\Delta U = Q - W$

$(°R - 491.67) \cdot 5/9 = °C$

@ $T_1 = 500°R = 4.63°C$

$\rightarrow u_1 = 85.20\ Btu/lb$

$T_2 = 700°R$

$\rightarrow u_2 = 119.58\ Btu/lb$

$$Q_{1-2} = U_{1-2} + W_{1-2} = 1\cancel{lb}\,(119.58 - 85.20)\ Btu/\cancel{lb} + 13.71\ Btu = 48.09\ Btu$$

$T_3 = 820°R$

$\rightarrow u_3 = 140.47\ Btu/lb$

$W_{3-1} = 0\ Btu$ b/c $\Delta V = 0$, therefore $W = 0$

$Q_{3-1} = U_{3-1} + W_{3-1}$

$Q_{3-1} = m(u_1 - u_3) + 0$

$Q_{3-1} = 1\cancel{lb}\,(85.2 - 140.47)\ Btu/\cancel{lb}$

$Q_{3-1} = -55.27\ Btu$

In adiabatic process, heat transfer = 0 Btu

$Q_{2-3} = U_{2-3} + W_{2-3}$

$0 = m(u_3 - u_2) + W_{2-3}$

$W_{2-3} = -m(u_3 - u_2)$

$= -(1\cancel{lb})(140.47 - 119.58)\ Btu/\cancel{lb}$

$W_{2-3} = \underline{-20.89\ Btu}$

$W_{1-2} = 13.71\ Btu$ $\quad Q_{1-2} = 48.09\ Btu$

$W_{2-3} = -20.89\ Btu$ $\quad Q_{2-3} = 0\ Btu$

$W_{3-1} = 0\ Btu$ $\quad Q_{3-1} = -55.27\ Btu$